Whirlpooling

From German brewing and more

Whirlpooling employs 2 methods of separating the trub from the wort. The first one is sedimentation, which means the trub will sink to the bottom when left undisturbed. The second one is centrifugal force which forces the trub into the center of the pot. If both methods are used, the trub will be collected in a nice trub-cone in the center of the pot. This is the main trub separation technique that is used in commercial brewing today before the wort is chilled. In a home brewing set-up, whirlpooling can be used before or after the wort is chilled. The latter also allows for the partial removal of cold break before the wort is transferred to the fermenter.

While sedimentation alone would work the trub would be evenly spread on the bottom of the pot preventing you from siphoning from a point as low as you could if the trub was collected in a cone.

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Whirlpooling After Wort Chilling

These are the steps for using whirlpooling after chilling the wort.

Chilling the wort

The wort is chilled with an immersion chiller. Note that the spoon has been left in the boil which sanitizes it for later use. If you don’t want to do that you can also sanitize the spoon with a sanitizer or use other sanitary means of starting a whirlpool (racking cane, oxygenation wand, turkey baster …)

Bring the wort into rotation

Then the pot is moved into an elevated position. The whirlpool can be started with a sanitized spoon or with an oxygenation wand. The latter is very practical since no spoon has to be sanitized and the oxygenation of the wort is already done during the whirlpool. After the whirlpool has been started it must not be moved or disturbed in order to allow for an even rotation of the wort. It is also important that there are no obstacles to the even rotation of wort in the kettle (e.g. fittings or pick-up tubes)

Rest the wort

The wort needs to rest for at least 20-30 min. Make sure it is covered. Now is also a good time to sanitize the racking and fermentation equipment.

Rack to the fermenter

Now that the trub has settled enough, the wort can be racked to a fermenter. The set-up shown here is more complicated than necessary as a post chiller has been added to get the wort temperature below 60*F (15*C). This chiller is coiled copper tubing in an ice bath. I have abandoned this method in favor of chilling the wort in the kettle to pitching temperatures using a pump to recirculate ice water from a bucket through the immersion chiller and back into the bucket.

When the racking cane is initially put in, it should not reach all the way to the bottom to avoid an early disturbance of the trub cone. As the wort level sinks the end of the racking cane can be lowered further. The siphon can be started with an auto siphon or even simpler with a turkey baster. Just squeeze the bulb of the turkey baster, attach the tip to the end of the hose and release the bulb. It will now suck the wort through the hose and start the siphon. Simply detach the turkey baster and let it flow.

If you are using a pot with a spigot, the spigot will have to sit higher than the trub for this to work.

As the wort level drops

As the wort level reaches the bottom and you peek into the pot you should start seeing the top of the trub cone. This cone will slowly spread as the wort level drops further. And when trub starts to get sucked up, it is time to stop the flow.

Collecting the wort from the trub

Since there will be a considerable amount of wort left in the trub it makes sense to collect it. If it is not needed for the primary fermentation (e.g. the post boil volume was lager than the volume that goes into the primary fermenter) there is no need to maintain sanitary conditions for this collection. A large funnel and paper towels work very well for filtering the wort from the trub-slush. Keep this wort frozen and use it for future starters or Kraeusening. But make sure it is boiled before use. If you have a pressure canner you can also pressure can the wort in Mason Jars.

This is the left over trub; a mix of hops, hot break and cold break

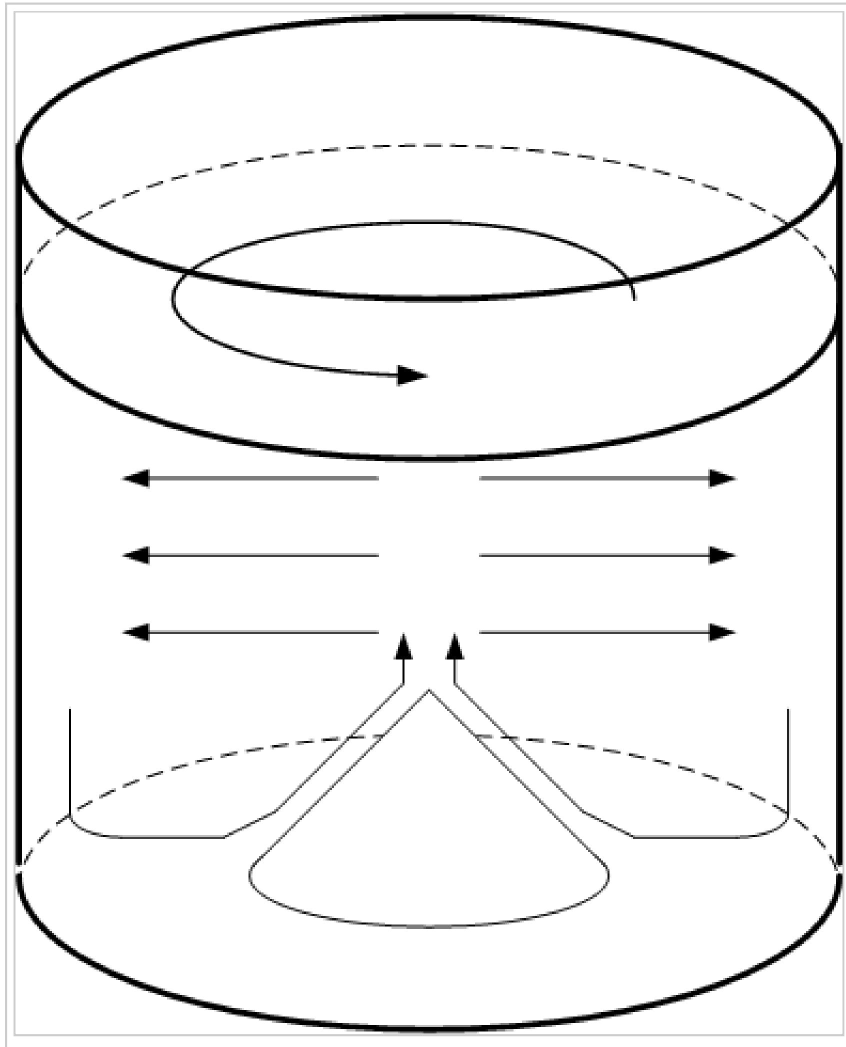
The result

This is the result: very little break material in the primary fermenter, which will make for easier yeast harvesting and a cleaner beer.

Whirlpooling before wort chilling

As mentioned earlier, whirlpooling can also be used before the wort is chilled. In this case the wort is chilled on the way to the fermenter by the use of a counterflow or plate chiller. It should be noted that all the cold break will end up in the primary fermenter. Cold break in the primary is not as concerning as hot break or hop material. There is however a concern regarding DMS. Since DMS is still formed as long as the wort is hot and its precursor (SMM) is present such DMS formation happens while the wort is resting in the whirlpool. Unlike during the boil, this DMS is not driven off and remains in the beer. This is especially a concern for beers made from very lightly kilned malts (Pilsner for example) which contain higher levels of SMM than darker kilned malts (Munich for example). If you are using a counterflow chiller to chill your wort and you are experiencing a DMS off-flavor in your beer, you may want to boils for longer or use an immersion chiller.

How does it work: Whirlpool dynamics



Why does the Whirlpool work the way it does? Shouldn't the centrifugal force move the tub to the outside and not to center of the pot?

Because of the rotation of the wort, wort is forced outward against the wall of the pot. This results in a pressure gradient, with pressure being lowest in the center and highest near the walls. Due to the friction between the bottom of the pot and the rotating wort, the wort closer to the bottom will not rotate as fast as the wort further up. This also means that its pressure gradient will be less. And since wort, like any other liquid or gas, moves from high pressure to low pressure there will be a spiraling inward motion of wort close to the bottom. The wort gets pretty much "sucked" into the center of the pot and is then forced outward again.

Friction on the wall and bottom of the pot causes the rotation to slow down. With that the inward current along the bottom gets slower as well. It is soon only able to move the break material into the center but not able to lift it up anymore - and thus the characteristic trub cone forms.

Since cold break is very fine, it is not really affected by the whirlpooling. Some settling of that material to the bottom occurs during the rest time.